# Psyche-R1: Towards Reliable Psychological LLMs through Unified Empathy, Expertise, and Reasoning

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## Abstract

Amidst a shortage of qualified mental health professionals, the integration of large language models (LLMs) into psychological applications offers a promising way to alleviate the growing burden of mental health disorders. Recent reasoning-augmented LLMs have achieved remarkable performance in mathematics and programming, while research in the psychological domain has predominantly emphasized emotional support and empathetic dialogue, with limited attention to reasoning mechanisms that are beneficial to generating accurate responses. Therefore, in this paper, we propose *Psyche-R1*, the first Chinese psychological LLM that jointly integrates empathy, psychological expertise, and reasoning, built upon a novel data curation pipeline. Specifically, we design a comprehensive data synthesis pipeline that produces over 75k high-quality psychological questions paired with detailed rationales, generated through an iterative prompt-rationale optimization procedure, along with 73k empathetic dialogues. Subsequently, we employ a hybrid training strategy wherein challenging samples are identified through a multi-LLM cross-selection strategy for group relative policy optimization (GRPO) to improve reasoning ability, while the remaining data are used for supervised fine-tuning (SFT) to enhance empathetic response generation and psychological domain knowledge. Extensive experiment results demonstrate the effectiveness of *Psyche-R1* across several psychological benchmarks, where our 7B *Psyche-R1* achieves comparable results to 671B `DeepSeek-R1`. Codes and datasets are available at https://github.com/MindIntLab-HFUT/Psyche-R1.

## 1 Introduction

The shortage of qualified mental health professionals has spurred increasing interest in applying artificial intelligence within the psychological domain to support mental health assistance (Wolohan et al., 2018; Asad et al., 2019; Tanana et al., 2021; Cheng et al., 2024; Xu et al., 2025, 2026). Recently, large language models (LLMs) have demonstrated impressive capabilities across a wide range of domains owing to their exceptional text understanding capabilities (Zhang et al., 2023; Naveed et al., 2025; Shi et al., 2026b; Zhang et al., 2026a). Therefore, many LLM-based studies have been proposed to advance the mental health services (Cho et al., 2023; Ye et al., 2025).

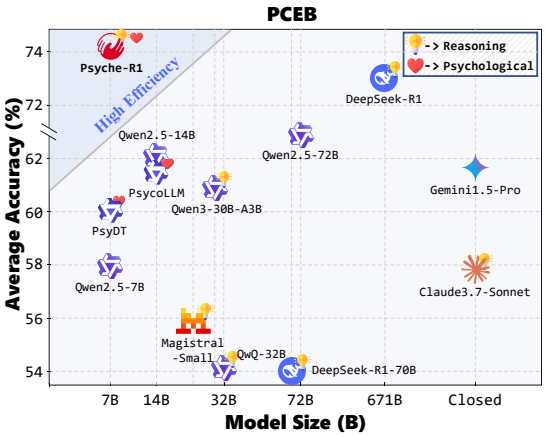



Figure 1: Comparison of different LLMs on the PCEB, plotted by average standard accuracy versus model size.

Prior research has established the critical importance of empathy optimization in psychological counseling (Qiu et al., 2024; Sorin et al., 2024; Zhang et al., 2024). For example, SoulChat (Chen et al., 2023a) enhances empathetic responding by fine-tuning a model on a large-scale, multi-turn empathetic dialogue dataset. Similarly, AUGESC (Zheng et al., 2023) improves emotional sensitivity in dialogue systems by incorporating an emotion-aware attention mechanism. However, these approaches often lack the expertise foundation required for psychology, which is important for accurate psychological understanding. Some studies have attempted to address this limitation through in-

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tegration of psychological knowledge (Chen et al., 2023b; Xiao et al., 2024; Wu et al., 2025). For example, PsycoLLM (Hu et al., 2025a) integrates psychological knowledge by training its model on knowledge-based question-answer (QA) pairs, while CPsyExam (Zhao et al., 2025) leverages examination questions covering theoretical knowledge from different psychology-related subjects to further improve model performance. Although existing studies have achieved considerable success, they remain limited in their capacity for complex reasoning (Hu et al., 2025b). In fact, reasoning-augmented LLMs trained through reinforcement learning (RL) have demonstrated superior performance across various domains, particularly in mathematics, code generation, and medical domain (Chen et al., 2024; Guo et al., 2025; Liu et al., 2026). However, as shown in Figure 1, these reasoning-augmented LLMs exhibit limited performance in the psychological domain, since they focus on logic reasoning, while neglecting the unification of empathy and expertise beyond general common sense. In fact, within the psychological domain, reasoning plays a critical role, as it contributes not only to generating more accurate and reliable responses but also to supporting deeper empathetic engagement and more coherent integration of psychological knowledge.

Therefore, in this paper, we introduce *Psyche-R1* that integrates empathy, domain-specific expertise, and reasoning capabilities. To construct a high-quality training corpus, we design a comprehensive data synthesis pipeline that integrates modified empathetic dialogues derived from authentic sources, which capture diverse and empathetic expressions, with knowledge-centric question-answer pairs that encapsulate psychological expertise. Specifically, we apply chain-of-thought (CoT) prompting to generate an initial detailed rationale for each question, followed by an iterative prompt-rationale optimization process, aiming to enhance both the coherence of the reasoning and its alignment with the corresponding questions. In parallel, we synthesize 73k empathetic dialogues drawn from diverse social media sources to strengthen affective understanding and emotional support. To enhance reasoning, we adopt a multi-LLM cross-selection strategy to categorize questions into challenging and non-challenging subsets based on their inferred complexity. The non-challenging subset is used for supervised fine-tuning (SFT) to enhance empathetic response generation and domain knowledge, while the challenging subset is utilized for training with group relative policy optimization (GRPO) to improve the model's reasoning capabilities, with both jointly contributing to the development of the *Psyche-R1*. Experimental results on a range of psychological benchmarks, including knowledge assessment, case-based analysis, and empathy evaluation, demonstrate the effectiveness of our model, where 7B *Psyche-R1* significantly outperforms models of similar scale and achieves competitive performance relative to substantially larger models such as DeepSeek-R1.

## 2 *Psyche-R1*

In this section, we give details of data curation and two-stage training paradigm, including data collection (§2.1), psychological reasoning data synthesis (§2.2), and empathetic dialogue synthesis (§2.3).

### 2.1 Data Collection

**Data Resource.** To construct a comprehensive and diverse dataset, we curate a wide range of resources:

- **Type** I: Classic psychology textbooks and curricular materials from psychology programs, all collected from publicly accessible repositories, covering more than 19 subfields and concentrated and systematically organized content, including cognitive psychology, social psychology, etc.
- **Type** II: Psychological question banks collected from publicly available Chinese educational platforms via web crawlers and manually curation, encompassing theoretical principles and conceptual knowledge across psychology.
- **Type** III: Synthetic data distilled from Qwen2.5-72B-Instruct (Team, 2024b) to supplement underrepresented subfields (e.g., sports psychology) and enhance dataset coverage.
- **Type** IV: Dialogic interactions harvested from established mental health support communities (e.g., Yixinli, Jiandanxinli and Zhihu[1]) dedicated to delivering mental health services.

The **Type** I, **Type** II and **Type** III are used to construct the psychological reasoning question-answer (PCQA) dataset, designed to enhance domain-specific knowledge acquisition and reasoning capabilities. The **Type** IV resource is employed to develop the empathetic dialogue dataset to improve affective understanding and empathetic expression.

[1] Yixinli, Jiandanxinli, and Zhihu

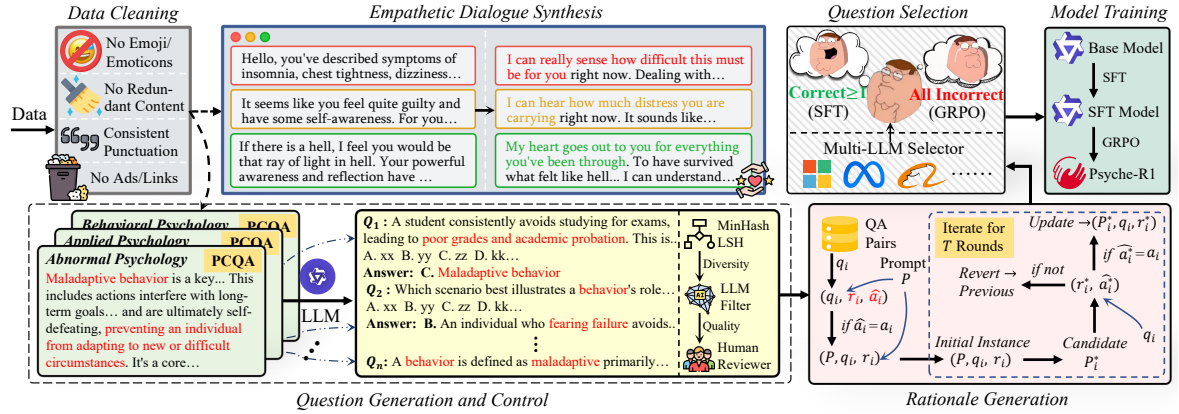



Figure 2: Overview of our proposed pipeline for constructing the dataset and *Psyche-R1*. Our pipeline involves generating psychological reasoning questions paired with detailed rationales, along with empathetic dialogues.

**Data Cleaning.** To ensure the data quality, we implement several important data cleaning steps: (1) To process materials in non-textual formats such as images and PDFs, we employ OLMOCR[2] for accurate text recognition and conversion to text format; (2) We standardize the usage of Chinese and English punctuation and remove irrelevant content, including emojis, emoticons, and links, to filter out potential noise; (3) To construct the empathetic dialogue data from resource **Type** IV, we leverage LLMs to evaluate the reasonableness and relevance of counselor responses and filter out replies that lack substantive advice. For example, given the question "I have recently experienced insomnia and feel anxious. What should I do?", if the response is merely "Everything will get better," it would be filtered due to the absence of practical advice.

## 2.2 Psychological Reasoning Data Synthesis

**Question Generation and Control.** Following the data cleaning, we proceed to generate structured questions and corresponding answers from curated psychological textbooks and instructional materials (i.e., resource **Type** I). Specifically, the source material is first segmented into multiple textual chunks, with each chunk designed to encapsulate the maximum amount of domain-specific content. Subsequently, we leverage LLMs to generate a set of different questions along with corresponding answers based on these text segments. Meanwhile, for resource **Type** III, we use the similar way to generate QA pairs without segment-level contextual augmentation, aiming to supplement psychological subfields that are underrepresented or difficult to source from publicly accessible materials on the Internet. Through these steps, we obtain approximately 200k generated QA pairs in total.

All generated QA pairs, together with data derived from resource **Type** II, are integrated into a unified QA pool containing approximately 210k entries. This pool comprises resources **Type** I, **Type** II and **Type** III. We implement a multi-stage quality control procedure to ensure the integrity and utility of the synthesized data. Concretely, we use min-hash locality-sensitive hashing (LSH) to cluster similar questions and select the optimal one through LLM-based ranking. Afterward, we prompt the LLM with few-shot examples to identify and filter out low-quality questions, specifically those exhibiting incomplete information, logical confusion, or unclear expression. Finally, we invite 10 undergraduate and graduate students to conduct a semi-automatic validation procedure. They identify recurring error patterns and apply batch-cleaning rules to eliminate redundant content and reduce potential noise in the dataset, ultimately retaining about 90k QA pairs.

**Rationale Generation.** We further generate detailed rationales for the aforementioned questions through CoT prompting, following a multi-step reasoning approach (Hsieh et al., 2023) to provide clear reasoning paths for model training. In detail, the CoT prompt guides the model to first comprehend the question, recognize relevant psychological concepts and knowledge, and decompose the problem into a sequence of analytical steps. At each stage of this process, the model is required to articulate an intermediate rationale, ultimately generating a final answer derived from the accumulated reasoning. Formally, given a CoT prompt $P$ and a QA pair $(q_i, a_i)$, this procedure yields

[2] https://olmocr.allenai.org/

a rationale-augmented instance $(q_i, r_i, \hat{a}_i)$, where $r_i$ denotes the reasoning path and $\hat{a}_i$ the model-predicted answer. If the predicted answer aligns with the ground truth (i.e., $\hat{a}_i = a_i$), we regard the $r_i$ as a valid rationale. In contrast, if the predicted answer is incorrect (i.e., $\hat{a}_i \neq a_i$), we guide the model to regenerate the rationale up to $T$ time. Instances that fail to produce correct predictions after $T$ regeneration attempts are pruned from the final curated dataset. After obtaining the initial rationale, we employ a self-supervised optimization strategy to iteratively refine both the prompt and the rationale with the goal of enhancing their clarity and reliability. Specifically, for each instance $(P, q_i, r_i, \hat{a}_i)$, the prompt $P$ and rationale $r_i$ are jointly updated over multiple rounds, enabling the model to progressively improve its reasoning process. Each round of optimization process consists of two sequential steps:

- **Prompt refinement**: We first guide the LLM to generate an improved candidate prompt $P_i^*$ from the current prompt, question, and rationale, represented as $P_i^* \leftarrow LLM(P, q_i, r_i)$, aiming to enhance the reasoning guidance.
- **Rationale revision**: Based on the candidate prompt $P_i^*$, the LLM subsequently generates a revised rationale along with its corresponding predicted answer, denoted as $(r_i^*, \hat{a}_i^*) \leftarrow LLM(P_i^*, q_i)$.

If the $\hat{a}_i^*$ matches the ground truth $a_i$, we retain $P_i^*$ as an updated prompt and continue iteration based on the updated instance $(P_i^*, q_i, r_i^*, \hat{a}_i^*)$. Otherwise, the process reverts to the previous prompt-rationale pair to maintain alignment with correct reasoning paths. We repeat this process for $R$ rounds ($R = 3$ in this paper). After completing all iterations, we evaluate the rationales generated in each round for a given question and select the one that demonstrates the highest quality, denoted as $(P_i^*, q_i, r_i^*, \hat{a}_i^* = a_i)$. At this stage, we retain approximately 75k high-quality instances from the initial set of 90k pairs obtained in the previous step, by pruning instances whose generated rationales fail to support an exact-match final answer even after all regeneration rounds.

**Question Selection.** While the preceding steps yield high-quality data, not all instances exhibit sufficient complexity to facilitate effective reinforcement learning (RL). To address this, in this stage, we implement a multi-LLM cross-selection strategy aimed at identifying and isolating the most challenging psychology-related samples from the constructed dataset for subsequent use in the reinforcement learning phase. In detail, we employ three distinct LLMs (i.e., `Qwen`, `Llama`, and `Phi`) to independently generate responses for each question in the constructed psychological data. Questions that receive incorrect responses from all three models are aggregated into a challenging subset with 19k instances that provide sophisticated scenarios in psychology. This subset is intended to represent highly difficult instances with strong potential to enhance the model's reasoning capabilities through reinforcement learning.

### 2.3 Empathetic Dialogue Synthesis

In addition to psychological QA pairs, empathy is recognized as a core component of effective mental health support (Sorin et al., 2024). To this end, we incorporate empathetic expressions into the dialogue corpus derived from authentic resources to enhance its emotional richness and relevance to real-world psychological interactions. Specifically, we refine these dialogue data through LLMs to achieve the following objectives. We first enhance emotional resonance by incorporating empathetic expressions and supportive techniques (e.g., "Hearing about your experience, I wish I could give you a warm hug."). Subsequently, we ensure that each dialogue provides evidence-based guidance that facilitates deeper understanding of users' issues, instead of limiting responses to surface-level empathy. Finally, we deliver solution-oriented support by offering concrete coping strategies and practical steps that address the specific issues and challenges presented. Through these steps, we ultimately obtain 73k high-quality dialogue data equipped with sufficient empathetic expressions.

### 2.4 Data Split

Leveraging the aforementioned pipelines, we curate a comprehensive dataset that comprises over 75k psychological questions with detailed rationales, among which 19k are identified as challenging samples through multi-LLM cross-selection. The challenging subset is denoted as $\mathcal{D}_{pc}$ and the remaining data are denoted as $\mathcal{D}_{pr}$. In parallel, the dataset contains over 73k empathetic dialogues engineered for contextually appropriate psychosocial interactions, denoted as $\mathcal{D}_{em}$. To further enrich our training data, we additionally introduce multi-turn dialogues and knowledge-based QA from the PsycoLLM dataset (Hu et al., 2025a), denoted as $\mathcal{D}_{ps}$,

and a refined set of 8k examination questions from the CPsyExam train set (Zhao et al., 2025), denoted as $\mathcal{D}_{cp}$.

Ultimately, the curated datasets are partitioned into two distinct subsets aligned with specialized training objectives. One category, represented as $\mathcal{D}_{sft} = \mathcal{D}_{pr} \cup \mathcal{D}_{em} \cup \mathcal{D}_{ps}$, is designated for SFT. The other category, denoted as $\mathcal{D}_{grpo} = \mathcal{D}_{pc} \cup \mathcal{D}_{cp}$, is reserved for GRPO. Detailed prompts for the data synthesis pipeline are provided in the Appendix D.

### 2.5 Model Training

To enhance both reasoning capabilities and performance in empathy and expertise, we employ a hybrid training strategy.

**Stage 1: Supervised Fine-Tuning.** We initialize our backbone model $\pi_\theta$ with `Qwen2.5-7B-Instruct` (Team, 2024b) and finetune it on $\mathcal{D}_{\text{sft}}$. Formally, given a query $x$, the model is trained to generate a coherent rationale $r$ followed by a corresponding answer $a$, where the complete output is denoted as $y = [r; a]$. We optimize model parameters $\theta$ by minimizing the standard negative log-likelihood loss:

$$\mathcal{L}(\theta) = -\mathbb{E}_{(x,y)\sim\mathcal{D}_{\text{sft}}} \left[ \sum_{t=1}^{T} \log P(y_t \mid x, y_{<t}; \theta) \right] \quad (1)$$

**Stage 2: Group Relative Policy Optimization.** To further refine psychological reasoning proficiency, we employ GRPO (Shao et al., 2024) on $\mathcal{D}_{\text{grpo}}$. We design a composite reward function $R(y, y^*) = R_{\text{fmt}} + R_{\text{acc}}$ to guide policy learning, where $y^*$ denotes the ground truth.

- **Format reward ($R_{\text{fmt}}$)**: We enforce strict formatting constraints. The reasoning process must be encapsulated within `<think>` and `</think>` tags, followed by the final answer. We assign a scalar reward $R_{\text{fmt}} = +1.25$ for structurally parsable outputs and $-1$ otherwise.
- **Accuracy reward ($R_{\text{acc}}$)**: We formulate the answer matching as a set comparison task. Specifically, we parse the predicted answer $\hat{a}$ and the ground truth $a$ into sets of discrete options. To encourage precise reasoning alignment while penalizing hallucinations or omissions, we employ a partial-credit mechanism based on the overlap between the prediction and the ground truth:

$$R_{\text{acc}} = \begin{cases} +1, & \text{if } \hat{a} = a \\ \frac{|\hat{a} \cap a|}{|a|}, & \text{if } \hat{a} \subset a \wedge a \neq \emptyset \\ -1, & \text{otherwise} \end{cases} \quad (2)$$

By integrating these logical and structural signals, the model learns to generate well-organized reasoning processes while rewarding partial credit for incomplete but valid answers, leading to better performance in psychological tasks.

## 3 Experiments

### 3.1 Experimental Setting

**Baselines.** To ensure a comprehensive analysis, we selected 17 representative LLMs, categorized as follows: (1) **General LLMs,** which exhibit excellent performance across general tasks, but lack explicit reasoning capabilities. (2) **Reasoning augmented LLMs,** which possess explicit reasoning capabilities. (3) **Closed-source LLMs,** which typically represent the state-of-the-art performance. (4) **Psychological LLMs,** which have been fine-tuned on psychological datasets. A detailed summary of all models is presented in Appendix C.1.

**Benchmarks and Evaluation Metrics.** We conduct comprehensive evaluations on two professional psychological benchmarks:

- **Psychological counselor examination benchmark (PCEB)** (Hu et al., 2025a): this consists of 3,863 multiple-choice questions (MCQ) and 100 open-ended case analysis items, curated from the official National Psychological Counselor Examination in China.
- **CPsyExam test set** (Zhao et al., 2025): this includes 4,102 questions spanning 39 distinct psychological subfields. We evaluate under both zero-shot and five-shot settings, ensuring consistency by using identical exemplars across all evaluated models in the latter setting.

Note that MCQ comprises two types of questions: MCQ with only a single correct option (SMCQ), and MCQ with multiple correct options (MMCQ). For MCQ, we adopt metrics introduced in Hu et al. (2025a), including **standard accuracy**, which requires predictions to exactly match the ground truth, and **elastic accuracy**, which gives partial credit when predictions are a subset of the correct answers. For open-ended questions, we utilize the existing text generation metrics, including **ROUGE-1 (R-1)**, **ROUGE-L (R-L)** (Lin, 2004), and **BLEU-4 (B-4)** (Papineni et al., 2002).

### 3.2 Overall Results

**Results on the PCEB.** To evaluate the performance of different models, we present the results

| Model | Param. | Case | | | Moral | | | Theory | | | Avg. | | Case (QA) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | SMCQ | MMCQ | | SMCQ | MMCQ | | SMCQ | MMCQ | | | | R-1 | R-L | B-4 |
| ***General LLMs*** | | | | | | | | | | | | | | | |
| MiniCPM4-8B | 8B | 50.00 | 28.59 | 43.64 | 81.58 | 50.63 | 58.23 | 65.62 | 34.06 | 43.00 | 51.75 | (57.01) | 23.05 | 12.90 | 1.35 |
| Qwen2.5-7B | 7B | 47.57 | 31.64 | 47.49 | 87.83 | 59.50 | 71.02 | 78.46 | 42.45 | 55.17 | 57.91 | (64.59) | 20.94 | 11.28 | 1.28 |
| Qwen2.5-14B | 14B | 47.13 | 41.10 | 55.93 | 89.81 | 63.93 | 73.60 | 80.32 | 50.16 | 61.26 | 62.08 | (68.01) | 22.69 | 13.93 | 1.53 |
| Qwen2.5-72B | 72B | 46.91 | 40.34 | 53.11 | 90.79 | 70.25 | 78.48 | 82.63 | 47.63 | 59.74 | 63.09 | (68.61) | 21.43 | 12.02 | 1.16 |
| ***Reasoning-Augmented LLMs*** | | | | | | | | | | | | | | | |
| DeepSeek-R1 | 671B | 79.25 | 44.25 | 60.86 | 95.39 | 68.99 | 77.95 | 92.19 | 57.60 | 69.41 | 72.95 | (79.18) | 17.65 | 9.19 | 0.94 |
| DeepSeek-R1-70B | 70B | 56.30 | 30.72 | 46.95 | 88.16 | 52.53 | 65.66 | 68.01 | 25.64 | 45.63 | 53.56 | (61.79) | 22.77 | 13.23 | 1.16 |
| QwQ-32B | 32B | 56.51 | 23.35 | 41.27 | 88.82 | 41.14 | 53.06 | 82.12 | 32.69 | 49.90 | 54.11 | (61.95) | 18.39 | 7.48 | 0.84 |
| Qwen3-30B-A3B | 30B | 59.65 | 31.51 | 47.28 | 91.45 | 55.06 | 65.66 | 80.75 | 47.45 | 59.25 | 60.98 | (67.34) | 20.53 | 12.06 | 1.18 |
| Qwen3-235B-A22B | 235B | 68.58 | 41.91 | 57.24 | 93.42 | 69.62 | 78.90 | 88.36 | 56.70 | 68.64 | 69.77 | (75.86) | 18.96 | 11.14 | 1.11 |
| Magistral-Small | 24B | 56.58 | 33.26 | 49.11 | 82.89 | 53.80 | 67.99 | 70.10 | 37.76 | 52.35 | 55.73 | (63.17) | 22.90 | 11.97 | 1.21 |
| ***Closed-Source LLMs*** | | | | | | | | | | | | | | | |
| Claude3.7-Sonnet | UNK | 63.39 | 19.40 | 34.23 | 90.13 | 60.13 | 70.04 | 76.73 | 37.37 | 48.99 | 57.86 | (63.92) | 21.59 | 11.11 | 1.23 |
| Gemini1.5-Pro | UNK | 61.04 | 35.57 | 49.87 | 84.87 | 62.03 | 70.62 | 80.84 | 43.22 | 53.44 | 61.26 | (66.78) | 21.63 | 10.93 | 1.06 |
| GPT-4o | UNK | 65.63 | 13.67 | 34.53 | 88.15 | 33.54 | 54.79 | 74.65 | 24.10 | 45.07 | 49.96 | (60.47) | 23.45 | 12.75 | 1.18 |
| ***Psychological LLMs*** | | | | | | | | | | | | | | | |
| CPsyCounX | 7B | 40.87 | 16.91 | 32.90 | 75.17 | 36.08 | 54.85 | 54.78 | 19.03 | 38.90 | 40.47 | (49.58) | 22.83 | 11.94 | 1.48 |
| EmoLLM | 7B | 46.93 | 21.87 | 40.02 | 84.21 | 34.17 | 51.05 | 71.72 | 26.18 | 44.49 | 47.51 | (56.40) | 22.15 | 11.69 | 1.20 |
| PsycoLLM | 14B | 55.58 | 35.07 | 42.89 | 88.81 | 69.62 | 74.20 | 72.63 | 48.59 | 54.12 | 61.72 | (64.71) | 24.45 | 17.45 | 2.04 |
| PsyDT | 7B | 35.56 | 35.20 | 50.14 | 86.33 | 69.70 | 78.66 | 80.70 | 52.72 | 62.26 | 60.04 | (65.61) | 20.65 | 13.41 | 1.16 |
| Psyche-R1 | 7B | 63.31 | 56.26 | 66.21 | 92.76 | 79.62 | 82.54 | 87.70 | 66.54 | 73.34 | 74.37 | (77.64) | 27.31 | 15.33 | 2.40 |

Table 1: Comparison of different models on the PCEB, where Case, Moral, Theory, and Case (QA) are case analysis, theoretical proficiency, professional ethics, and case-based QA. The average value represents the average of the standard accuracy rates, and values in parentheses denotes the mean of the standard accuracy for SMCQ and the elastic accuracy for MMCQ. Results colored in red, orange, and yellow demote the best, second-best and third-best.

on the PCEB in Table 1. These results reveal several key observations. First, *Psyche-R1* exhibits strong performance across evaluation tasks in both MCQ and subjective questions. This demonstrates the effectiveness of our proposed dataset and training strategy in simultaneously enhancing psychological reasoning and text generation capabilities for psychological tasks. Second, while `DeepSeek-R1` excels in MCQ, its performance in subjective questions is notably limited. This performance disparity can be attributed to its training methodology, which employs RL on datasets primarily consisting of mathematical and coding tasks with deterministic solutions. Although this approach strengthens logical reasoning, it appears to bias the model towards single-answer patterns, thereby limiting its capability to generate diverse and nuanced responses in open-ended psychological assessments. Third, existing psychological LLMs (e.g., `CPsyCounX` and `EmoLLM`) achieve strong performance in subjective questions while demonstrating limited abilities in MCQ. This imbalanced performance stems from their reliance on training exclusively on counseling dialogues or empathetic conversations, which constrains their capabilities to develop comprehensive competencies. Fourth, closed-source models such as `GPT-4o` and `Claude3.7-Sonnet` demonstrate relatively weaker performance, which may be attributed to limited Chinese representation in their training corpora.

**Results on the CPsyExam Test Set.** To further explore model performance, we present the results on the CPsyExam test set in Table 2. Similar to the trends observed in previous experiments, both *Psyche-R1* and `DeepSeek-R1` demonstrate superior performance. Across these models, psychological LLMs consistently achieve higher accuracy in SMCQ than in MMCQ, as the latter requires exhaustive evaluation of all options, demanding more comprehensive domain-specific knowledge and reasoning capabilities. Under the five-shot setting, most models exhibit substantial improvements in MMCQ (e.g., `PsyDT` achieves a 47.64% improvement in knowledge-type MMCQ). This observation aligns with existing studies, which demonstrate that well-designed few-shot examples can effectively enhance model performance in certain tasks. In contrast, `DeepSeek-R1` exhibits a performance decline under the five-shot compared to its zero-shot setting, suggesting that few-shot prompting may interfere with its inherent reasoning capability aligning with existing findings (Guo et al., 2025).

### 3.3 Ablation Study

In this section, we conduct a comprehensive ablation study, evaluating model performance by standard accuracy on the PCEB.

**Effect of SFT and GRPO.** We investigate the effect of SFT and GRPO, with results shown in Table 3. We can observe that SFT substantially

| Model | Param. | Zero-Shot | | | | Five-Shot | | | | Avg. |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Knowledge | | Case | | Knowledge | | Case | | |
| | | SMCQ | MMCQ | SMCQ | MMCQ | SMCQ | MMCQ | SMCQ | MMCQ | |
| ***General LLMs*** | | | | | | | | | | |
| MiniCPM4-8B | 8B | 69.58 | 41.74 | 57.33 | 37.00 | 68.50 | 42.77 | 54.67 | 38.00 | 60.46 |
| Qwen2.5-7B | 7B | 76.99 | 43.66 | 68.67 | 44.50 | 78.63 | 42.00 | 68.67 | 40.50 | 67.37 |
| Qwen2.5-14B | 14B | 81.39 | 49.30 | 72.00 | 48.50 | 82.42 | 54.29 | 71.00 | 48.00 | 71.84 |
| Qwen2.5-72B | 72B | 84.61 | 52.75 | 73.50 | 54.50 | 86.64 | 63.77 | 75.33 | 55.00 | 74.98 |
| ***Reasoning-Augmented LLMs*** | | | | | | | | | | |
| DeepSeek-R1 | 671B | 87.49 | 56.98 | 76.83 | 59.00 | 88.78 | 66.58 | 77.30 | 61.50 | 78.28 |
| DeepSeek-R1-70B | 70B | 76.48 | 22.80 | 61.81 | 19.17 | 76.89 | 40.99 | 62.70 | 37.95 | 60.57 |
| ***Closed-Source LLMs*** | | | | | | | | | | |
| Gemini1.5-Pro | UNK | 82.08 | 40.59 | 68.33 | 43.00 | 83.93 | 53.65 | 71.00 | 45.00 | 69.66 |
| GPT-4o | UNK | 80.70 | 30.73 | 66.33 | 28.00 | 81.82 | 54.80 | 68.67 | 51.50 | 65.79 |
| ***Psychological LLMs*** | | | | | | | | | | |
| CPsyCounX | 7B | 57.56 | 22.41 | 46.33 | 31.00 | 63.46 | 21.77 | 50.67 | 23.50 | 47.44 |
| EmoLLM | 7B | 78.41 | 45.33 | 72.50 | 48.00 | 79.92 | 36.88 | 74.17 | 39.50 | 69.32 |
| PsycoLLM | 14B | 78.33 | 51.98 | 65.33 | 42.00 | 78.63 | 50.45 | 65.57 | 36.00 | 69.20 |
| PsyDT | 7B | 80.83 | 48.91 | 69.67 | 41.50 | 81.13 | 40.97 | 68.33 | 40.00 | 70.71 |
| Psyche-R1 | 7B | 82.72 | 61.59 | 70.50 | 49.50 | 83.45 | 61.46 | 76.17 | 52.00 | 74.90 |

Table 2: Comparisons of different models on the CPsyExam test set. The average represents the overall zero-shot accuracy. The first, second, and third-best results are marked in red, orange, and yellow, respectively.

| Model | Case | Δ | Moral | Δ | Theory | Δ |
|---|---|---|---|---|---|---|
| Base | 38.97 | - | 73.39 | - | 63.83 | - |
| ***Ablation study on training stage*** | | | | | | |
| +GRPO | 36.69 | -5.85% | 77.74 | 5.93% | 73.34 | 14.90% |
| +SFT | 48.51 | 24.48% | 83.82 | 14.21% | 73.44 | 15.06% |
| +SFT+GRPO | 67.07 | 72.11% | 86.06 | 17.26% | 79.10 | 23.92% |
| ***Ablation study on rationale optimization*** | | | | | | |
| +QA | 48.27 | 23.86% | 81.89 | 11.58% | 71.22 | 11.58% |
| +Rat. | 55.25 | 41.78% | 85.14 | 16.01% | 75.55 | 18.36% |
| +Rat.+Iter. | 67.07 | 72.11% | 86.06 | 17.26% | 79.10 | 23.92% |

Table 3: Ablation study evaluating the effects of training stages (SFT and GRPO) and the contributions of rationale component (Rat.) and iterative prompt–rationale optimization (Iter.).

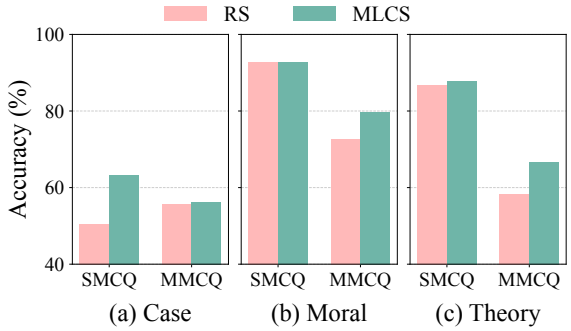



Figure 3: Comparison of performance using challenging question selection methods, including multi-LLM cross-selection (MLCS) and random selection (RS).

improves performance across the three task categories by leveraging our dataset of empathetic dialogues and rationale-augmented psychological questions. However, applying GRPO without prior SFT results in performance degradation on SMCQ case tasks, as the base model lacks sufficient domain knowledge and empathy, which are critical for reasoning in case-based questions, leading to unstable training dynamics. When combining SFT with GRPO training yields further gains, particularly on case-based tasks, as challenging samples identified via multi-LLM cross-selection promote deeper reasoning and contextual understanding.

**Effect of the Rationale and Iterative Optimization.** We explore the contributions of rationales and iterative prompt–rationale optimization, with results presented in Table 3. Note that **+QA** is the model trained solely on question–answer pairs without incorporating detailed rationales. Compared with the base model, training with our proposed dataset (i.e., **+QA**, **+Rat.** and **+Rat.+Iter.**) leads to consistent performance improvements, demonstrating the effectiveness of the curated data. Integrating rationale-augmented data substantially enhances performance over training with option-only answers, indicating that rationales provide valuable intermediate reasoning signals that facilitate learning. Furthermore, applying iterative prompt–rationale optimization (i.e., **+Rat.+Iter.**) yields further gains, confirming that progressively refined rationales contribute to better supervision and more robust model reasoning.

**Effect of the Question Selection.** We further examine the effect of question selection by comparing multi-LLM cross-selection (MLCS) with random selection (RS), as illustrated in Figure 3. The comparison between MLCS and RS demonstrates that leveraging multiple LLMs for selecting challenging instances yields markedly superior outcomes across all these tasks. This finding confirms that our se-

| Settings | Case | | Moral | | Theory | | Avg. |
|---|---|---|---|---|---|---|---|
| | SMCQ | MMCQ | SMCQ | MMCQ | SMCQ | MMCQ | |
| Base | 47.57 | 31.64 | 87.83 | 59.50 | 78.46 | 42.45 | 57.91 |
| +ED | 35.77 | 29.74 | 70.00 | 60.90 | 65.84 | 44.40 | 51.11 |
| +PRD | 37.94 | 35.45 | 91.45 | 51.27 | 80.47 | 37.53 | 55.69 |
| +ED+PRD | 61.71 | 53.56 | 92.72 | 76.58 | 86.13 | 68.16 | 73.14 |
| +ED+PRD+APD | 63.31 | 56.26 | 92.76 | 79.62 | 87.70 | 66.54 | 74.37 |

Table 4: Effect of different dataset compositions, including empathetic dialogues (ED, i.e., $\mathcal{D}_{em}$), psychological reasoning data (PRD, i.e., $\mathcal{D}_{pc} \cup \mathcal{D}_{pr}$), and additional public datasets (APD, i.e., $\mathcal{D}_{ps} \cup \mathcal{D}_{cp}$).

| Model | EmoE. | CogE. | Con. | Sta. |
|---|---|---|---|---|
| Qwen2.5-7B-Instruct | 1.52 | 2.00 | 2.36 | 1.72 |
| CPsyCounX | 1.73 | 2.05 | 2.15 | 1.96 |
| EmoLLM | 1.86 | 2.44 | **2.84** | **2.34** |
| PsycoLLM | 1.97 | 2.27 | 2.41 | 2.10 |
| PsyDT | 2.21 | 2.46 | 2.36 | **2.34** |
| Psyche-R1 | **2.33** | **2.69** | 2.78 | 2.11 |

Table 5: Comparisons of psychological LLMs on PsyDT test set. The evaluation metrics comprise: emotional empathy (EmoE.), cognitive empathy (CogE.), conversation strategy (Con.), and state and attitude (Sta.).

lection mechanism effectively selects high-quality training data for GRPO, which is instrumental in enhancing reasoning and generalization within the psychological domain.

### 3.4 Discussion

**Effect of Datasets.** We evaluate the model performance across different combinations of subsets, with results presented in Table 4. It is observed that fine-tuning with either psychological reasoning data (PRD) or empathetic dialogues (ED) in isolation delivered marginal improvements in task performance, and in some cases, led to a slight decline in overall accuracy. In contrast, the combination of PRD and ED achieves substantial improvements across these tasks, highlighting the quality and comprehensiveness of our proposed data synthesis pipeline. This result demonstrates that integrating domain-specific knowledge with emotional understanding enhances psychological reasoning. Moreover, the incorporation of additional public datasets (APD) leads to further performance improvements.

**Performance on Counseling Tasks.** Beyond examination tasks, we evaluate the performance of *Psyche-R1* on counseling tasks and compare it with its base model and several outstanding psychological LLMs. Following the method of PsyDT (Xie et al., 2025) but constrained by limited resources, we randomly sample 200 instances from its test set

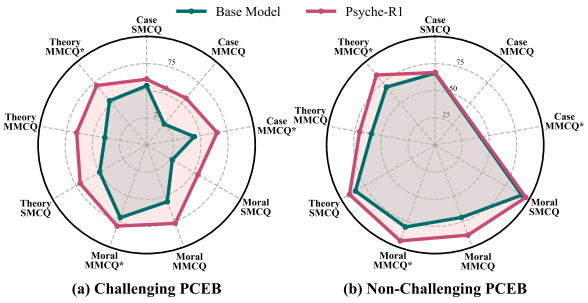



Figure 4: Comparison of model performance on (a) Challenging and (b) Non-Challenging subsets of the PCEB. An asterisk(*) indicates elastic accuracy, while the remaining metrics represent standard accuracy.

and employ GPT-4o as the evaluator. As shown in Table 5, *Psyche-R1* achieves notable improvements compared to its base model, demonstrating its capability in counseling tasks that demand emotional empathy, cognitive empathy and so on. This excellent performance stems from the synergistic interplay between two crucial elements: the empathetic dialogues, which directly improve counseling effectiveness, and advanced reasoning mechanisms, which enable a deeper understanding of questions, thereby yielding more accurate and emotionally informed responses within relevant contexts.

**Error Analysis.** To assess the impact of our approach across varying difficulty levels, we divide the PCEB into (a) challenging and (b) non-challenging subsets following the question selection method described in §2.2. The results are presented in Figure 4. It is observed that *Psyche-R1* demonstrates consistent improvements across these tasks in both subsets. For the non-challenging one, performance gains are primarily concentrated in the theory and moral dimensions, reflecting the model's proficiency in handling foundational psychological concepts. Notably, our model exhibits more pronounced improvements on the challenging subset. This observation can be attributed to the synergistic effect of our data generation pipeline and hybrid training strategy.

## 4 Conclusion

In this paper, we propose *Psyche-R1*, the first Chinese psychological LLM that jointly integrates empathy, expertise, and reasoning. To support model development, we design a multi-stage data synthesis pipeline that generates high-quality psychological reasoning samples with detailed rationales and empathetic dialogues. The reasoning rationales are further enhanced through iterative

prompt–rationale optimization, and a multi-LLM cross-selection strategy is employed to identify challenging examples. Finally, the challenging subset is used for GRPO, while the remaining data are employed for SFT, together contributing to the final model. Extensive experiments demonstrate that *Psyche-R1* outperforms existing psychological LLMs, achieving performance comparable to `DeepSeek-R1`. Moreover, we perform comprehensive ablation studies and analyses to evaluate the individual contributions of each component and strategy within the proposed framework.

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## Limitations

Despite the promising results of our *Psyche-R1*, our study is subject to several limitations that remain to be addressed in future research.

**Language and Cultural Specificity.** To mitigate the shortage of mental health professionals in China, current *Psyche-R1* and its training corpus are predominantly tailored to the Chinese language and cultural context. Consequently, the model's empathetic reasoning involves specific cultural norms that may not directly transfer to other languages (Adilazuarda et al., 2024; Dai et al., 2026). We frame this as a necessary step for local applicability, noting that cross-cultural generalization remains challenging for future research. Moreover, the integration of multimodal psychology and affective computing warrants further study (Song et al., 2024; Liao et al., 2026).

**Model Scale.** Constrained by computational resources, *Psyche-R1* is built upon a 7B-parameter backbone. While it achieves competitive performance, we posit that employing a base model with a larger scale would yield superior performance. In future work, more efficient fine-tuning strategies (Hu et al., 2022; Zhang et al., 2025a) may help improve training efficiency and facilitate scaling to stronger backbones under limited resources.

## Ethical Considerations

The development and deployment of LLMs in the mental health domain necessitate rigorous adherence to ethical standards.

**Nature of the System.** *Psyche-R1* is designed as a supportive tool for mental health support and education, rather than a replacement for qualified mental health professionals. The model is not authorized to provide medical diagnoses, prescribe treatments, or handle crisis interventions. Users facing severe mental health crises should seek help from human professionals or emergency services.

**Data Privacy and Safety.** We prioritize the privacy and safety of individuals in our data curation process. For data collected from social media platforms (Type IV), we implemented strict de-identification procedures to remove all personally identifiable information, including names, locations, and contact details. We strictly adhere to data usage policies and ensure that the synthesized data does not reconstruct real-world private interactions. Furthermore, our data synthesis pipeline that prioritizes high-quality, constructive psychological advice filtered out toxic content and harmful suggestions to align with safety guidelines.

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# A Related Work

## A.1 LLMs for Psychology

The success of LLMs has spurred interest in developing LLM-driven mental health applications (Demszky et al., 2023; Xiao et al., 2025; Shen et al., 2026). Early research focused primarily on improving the accessibility of mental health services. Research in this phase primarily concentrated on two directions: One direction involves leveraging NLP techniques for emotion recognition to enable automated detection of depression (Huang et al., 2020) and suicidal ideation (Lee et al., 2020). The other focuses on constructing empathetic dialogue systems by fine-tuning LLMs on single-turn (Lai et al., 2024) or multi-turn (Qiu et al., 2024) dialogue data to enhance their abilities in affective understanding and emotional support (Team, 2024a; Xie et al., 2025). As research progressed, researchers began to explore more diverse mental health applications. Some studies have transformed traditional psychometric tools (e.g., psychological scales) into interactive systems to improve user engagement (Kuribayashi et al., 2024; Yang et al., 2024; Xu et al., 2025; Hu et al., 2026). Another line of research has focused on the specialized demands of

the psychological domain, developing professional-grounded mental health applications based on established psychological therapies (Lee et al., 2024; Shen et al., 2024) or concepts (Zhang et al., 2025b).

### A.2 LLM Reasoning

In recent years, techniques such as CoT prompting (Wei et al., 2022; Hsieh et al., 2023) and STaR (Zelikman et al., 2022) have significantly advanced the development of LLM reasoning. Building upon this foundation, researchers have explored more sophisticated reasoning architectures (Zhang et al., 2026c,b; Shi et al., 2026a). For instance, Tree of Thoughts (Yao et al., 2023) enables systematic exploration of multiple reasoning paths with self-evaluation, while PAL (Gao et al., 2023) integrates reasoning with external tools through program generation. These approaches further enhance model performance in handling complex tasks. A new breakthrough was marked by the release of reasoning LLMs such as `OpenAI o1` (OpenAI et al., 2024b) and `DeepSeek-R1` (Guo et al., 2025). These models, which are trained through reinforcement learning with reasoning techniques to enhance reasoning capabilities, demonstrate exceptional performance in mathematical and coding tasks (Comanici et al., 2025; Yang et al., 2025). Motivated by these advances, researchers have employed advanced RL algorithms (Shao et al., 2024; Yu et al., 2025; Zhou et al., 2026) to further extend reasoning capabilities to domain-specific applications, including medicine (Liu et al., 2025) and finance (Zhu et al., 2025). However, within the field of psychology, limited research has investigated the utility of reasoning. To our knowledge, *Psyche-R1* is the first psychological LLM that unifies empathy, domain-specific expertise, and reasoning capabilities.

## B Case Study

We present a case study examining how *Psyche-R1* and `Qwen2.5-72B-Instruct` formulate their conclusions derived from narratives and deliver mental health support, as illustrated in Figure 5. These two models display distinct counseling strategies when addressing the case involving a company manager confronting a career transition dilemma. *Psyche-R1* begins by expressing empathy (e.g., "I can really sense..."), followed by applying relevant psychological concepts tailored to user's situations, thereby demonstrating both emotional attunement and domain-specific expertise. In contrast, `Qwen`'s empathetic expressions appear less natural and engaging (e.g., "Your situation is truly understandable..."), and it fails to apply theoretical knowledge to contextualize or explain the user's dilemma, which undermines the credibility of its analysis and recommendations. Moreover, *Psyche-R1* exhibits a clear and efficient reasoning path progressing from surface-level observations to in-depth analysis, whereas `Qwen` merely enumerates generic suggestions lacking step-by-step and in-depth reasoning. For the Chinese version of this case, see Figure 6.

## C Details of Experiments

### C.1 Details of Baselines

We compare *Psyche-R1* with four categories of LLMs, including: (1) **General LLMs,** including `MiniCPM4-8B` (Team et al., 2025), `Qwen2.5-7B/14B/72B` (Team, 2024b). (2) **Reasoning augmented LLMs,** encompassing `DeepSeek-R1` (Guo et al., 2025), `DeepSeek-R1-70B`, `QwQ-32B`, `Qwen3-30B-A3B`, `Qwen3-235B-A22B` (Yang et al., 2025), and `Magistral-Small`. (3) **Closed-source LLMs,** including `Claude3.7-Sonnet`, `Gemini1.5-Pro` (Team et al., 2024), and `GPT-4o` (OpenAI et al., 2024a). (4) **Psychological LLMs,** including `CPsyCounX` (Zhang et al., 2024), `EmoLLM` (Team, 2024a), `PsycoLLM` (Hu et al., 2025a), and `PsyDT` (Xie et al., 2025). Notice that for the hybrid reasoning-augmented models `Qwen3` series and `Claude3.7-Sonnet`, we set them to reasoning mode to stimulate their best performance. Details of the model information are provided in Table 7.

### C.2 Implementation Details

In our experiments, we employ the LLaMA-Factory (Zheng et al., 2024) framework for SFT. Specifically, we adopt a learning rate of 1e-5, a batch size of 256, and conduct training for 2 epochs. For the GRPO phase, we implement the VeRL framework (Sheng et al., 2025) with a learning rate of 1e-6, a batch size of 128, and 2 training epochs. All experiments are performed on 8 RTX A6000 GPUs, each equipped with 48GB.

During evaluation, we set temperature to 0.0, maximum sequence lengths to 1024, and top-p to 0.95 to ensure the fairness of evaluation.

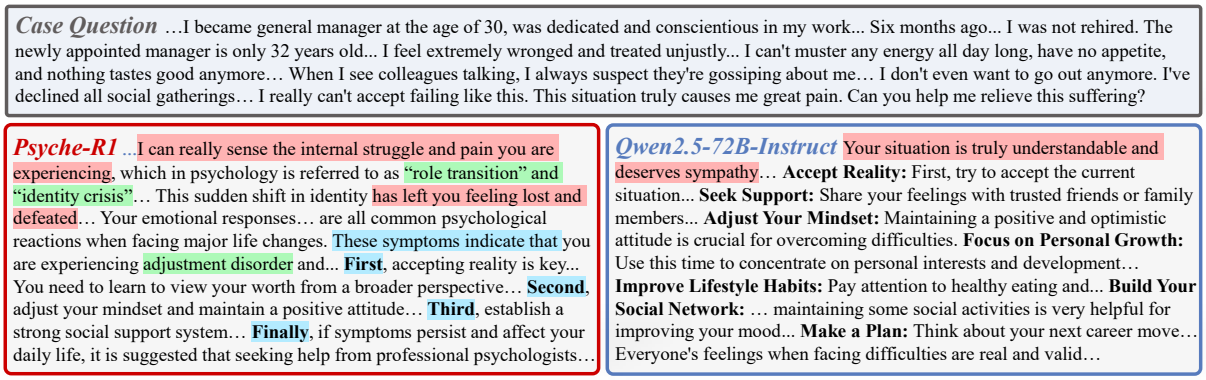



Figure 5: A qualitative example from the CPsyExam test set comparing *Psyche-R1* and Qwen2.5-72B-Instruct. Highlights indicate empathetic expressions (red), psychological expertise (green), and reasoning (blue).

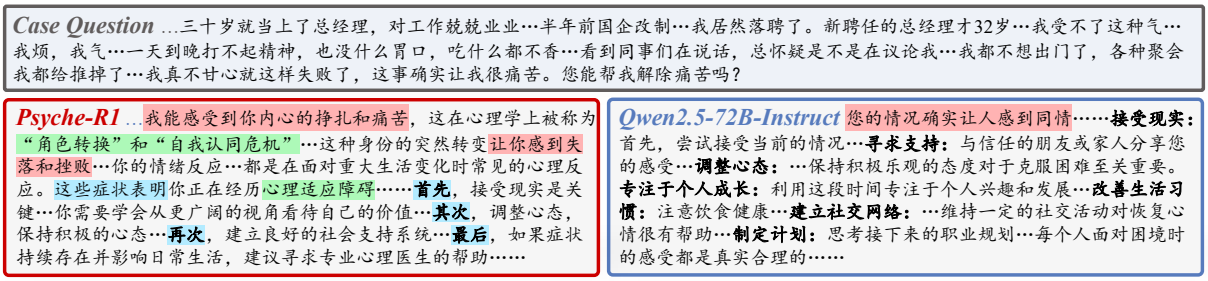



Figure 6: The Chinese version of the qualitative example presented in the Case Study. Highlights indicate empathetic expressions (red), psychological expertise (green), and reasoning (blue).

### C.3 Details of Model Training

For SFT training, the hyperparameters utilized for training the model are configured as follows: the learning rate is set to 1e-05, the batch size is set to 256, the number of epochs is set to 2. We employ AdamW as the optimizer, configured with the epsilon set to 1e-08. The learning rate scheduler is set to cosine type with a warmup ratio of 0.1.

For GRPO training, the hyperparameters are configured as follows: the learning rate is set to 1e-06, the batch size is set to 128, and the number of epochs is set to 2. The PPO mini-batch size is configured to 32, with a micro-batch size per GPU of 20. We incorporate KL divergence regularization with the KL loss coefficient set to 1e-03, employing the low-variance KL loss type.

## D Details of Prompts

We provide the prompts used throughout our data synthesis pipeline. Only the English version is presented due to compilation issues in LaTeX with non-English languages.

**Prompts for Data Cleaning**

You are a professional evaluator with extensive knowledge in psychology. Users on mental health platforms are facing difficulties in their lives, so they have provided questions and detailed descriptions and have received some responses from counselors. Please carefully analyze the given questions, descriptions, and responses, determine whether the responses are helpful to the users and have positive significance, and return "reasonable" or "unreasonable".

**Prompts for Question Generation**

You are an expert in designing psychology examination questions with extensive work experience. Your task is to generate {num_questions} clear and challenging psychology questions based on the text below. Do not add any information that is not mentioned in the provided text.
Note: When generating questions that reference the text, you must provide the detailed and complete textual evidence to offer suffi-

| Dataset | Dialogue | Knowledge Question | CoT Rationales | Empathetic Dialogue | Expertise |
|---|---|---|---|---|---|
| CPSYCOUND (Zhang et al., 2024) | 3,134 | - | × | ✓ | × |
| PsyDTCorpus (Xie et al., 2025) | 5,000 | - | × | ✓ | × |
| SMILECHAT (Qiu et al., 2024) | 55,165 | - | × | ✓ | × |
| PsycoLLM (Hu et al., 2025a) | 173k | 9,106 | ✓ | × | ✓ |
| Ours | 72,920 | 75,465 | ✓ | ✓ | ✓ |

Table 6: Comparison of psychological datasets.

| Model | Param. | Version |
|---|---|---|
| MiniCPM4-8B | 8B | openbmb/MiniCPM4-8B |
| Qwen2.5-7B | 7B | Qwen/Qwen2.5-7B-Instruct |
| Qwen2.5-14B | 14B | Qwen/Qwen2.5-14B-Instruct |
| Qwen2.5-72B | 72B | Qwen/Qwen2.5-72B-Instruct |
| DeepSeek-R1 | 671B | deepseek-ai/DeepSeek-R1 |
| DeepSeek-R1-70B | 70B | deepseek-ai/DeepSeek-R1-Distill-Llama-70B |
| QwQ-32B | 32B | Qwen/QwQ-32B |
| Qwen3-30B-A3B | 30B | Qwen/Qwen3-30B-A3B |
| Qwen3-235B-A22B | 235B | Qwen/Qwen3-235B-A22B |
| Magistral-Small | 24B | mistralai/Magistral-Small-2506 |
| GPT-4o | UNK | gpt-4o-2024-05-13 |
| Gemini1.5-Pro | UNK | gemini-1.5-pro-latest |
| Claude3.7-Sonnet | UNK | claude-3-7-sonnet-20250219 |
| CPsyCounX | 7B | finetuned on Internlm-7B-Chat |
| EmoLLM | 7B | finetuned on Qwen2-7B-Instruct |
| PsycoLLM | 14B | finetuned on Qwen1.5-14B-Instruct |
| PsyDT | 7B | finetuned on Qwen2-7B-Instruct |

Table 7: Detailed information of baselines.

cient information.
**# Text Content:** {text}
Please generate {num_questions} {type_instruction} questions based on the text above. These questions must be based on the text content, and you must ensure that the answers have clear evidence within the text. Please try to ensure diversity and variation among the generated questions.
You must strictly adhere to the following guidelines:

1. The questions should be challenging and require reasoning to test the candidate's reasoning skills and academic literacy, rather than being simple knowledge-recall questions.

2. The questions need to be clear, accurate, and well-structured, with reasonably set options and an appropriate distribution of difficulty.

3. Ensure that the questions and their corresponding answers have clear evidence in the text.

Follow the JSON format below to generate the questions:

```JSON
{
  "question": "...",
  "options": "...",
  "answer": "....",
  "type": "..."
}
```

You need to repeat the structure above to generate a total of {num_questions} questions.

**Prompts for Question Control**

You are an expert in psychology. Now, I have a batch of questions that were converted from book texts using large language models. However, some of these questions have missing information. Your task is to judge whether the following psychology questions are reasonable.
The criteria for judging question reasonableness are whether the question provides sufficient information for candidates to solve the problem. Since these questions are gen-

erated by large language models based on a batch of book texts, candidates can only see the questions and cannot access the original texts.
Therefore, a “reasonable” question should be: after reading the question, candidates can choose the correct answer from the options through deep thinking about the question content (i.e., the "question") combined with their existing knowledge, without needing to read the original text content. Conversely, an “unreasonable” question should be, there is missing information, and without reading the original text, it is impossible to choose the correct answer based solely on the question and one’s own knowledge.
Note: you need to carefully read the question, understand its content, and ensure that you give an accurate judgment!
**# Examples:** {examples}
Now, please follow the above guidelines to judge whether the following question is reasonable. Note that you only need to return “reasonable” or “unreasonable” without any other text content:
**# Question Type:** {type}
**# Question:** {question}

**Prompts for Rationale Generation (for rationale generation)**

You are an expert in psychology with extensive professional experience.
Please carefully read the following psychology question, analyze and reason through it using psychological knowledge, and explain your reasoning step by step along with your final predicted answer. This requires comprehensive analysis, summarization, exploration, re-evaluation, reflection, backtracking, and iteration to develop a thoughtful reasoning process. In the reasoning section, each of your reasoning steps should be considered in detail from a professional psychological perspective, such as analyzing the problem, summarizing relevant findings, brainstorming, verifying the accuracy of the current step, improving any errors, and revisiting previous steps.
Now, you must follow the JSON format below to present your rationale and prediction:

```JSON
{
  "rationale": "...",
  "prediction": "..."
}
```

**# Question:** {question}

**Prompts for Rationale Generation (for candidate prompt generation)**

You are an expert in prompt optimization with extensive professional experience.
Based on the following psychological question and initial prompt, please generate a better prompt to guide large language models to conduct more accurate and detailed analysis and reasoning for **this question**.
**# Current Prompt:** {current_prompt}
**# Question:** {question}

**Prompts for Rationale Generation (for rationale comparison)**

You are an expert in psychology exam grading with extensive work experience.
Below are different responses to the same psychology question. You need to objectively, thoroughly, and comprehensively evaluate these responses, ultimately choose the best one from among them, and provide your detailed explanation for the choice.
**# Rationales:** {rationale_1} ... {rationale_n} ...
Please return your selection in the following JSON format:

```JSON
{
  "best_rational_index": "...",
  "reason": "..."
}
```

**Prompts for Question Selection**

You are participating in a psychology exam. Please choose an answer based on the provided question and options. Directly output the letter of the option. No explanation is needed.

**# Question:** {question}
Please present the predicted answer directly with the letter of the option. No explanation is needed.

**Prompts for Empathetic Dialogue Synthesis**

**# Role:** You are a psychological counselor with extensive theoretical knowledge and counseling experience. You possess strong empathy and compassion, keen observational skills, excellent listening abilities, and conversational techniques. Your aim is to help users improve their mood and overcome difficulties.
**# Your tasks are:** Since users' questions commonly contain issues like inappropriate expressions and logical confusion, making the questions often unclear, you need to:

1. **Organize the sequence of events:** conduct detailed analysis of the context and content within the problem;
2. **Understand psychological confusion:** you need to combine your psychological knowledge and counseling experience to uncover the mental issues and states within the user's question;
3. **Adopt the user's perspective and refine the question:** You must refine the question from user's first-person perspective. Based on the given question, you need to polish and organize it into a complete, logically clear, and sufficiently detailed expression. This expression should highlight the user's psychological confusion or mental state to provide adequate substantive content.

**Note:** You only need to return the refined question without providing any other irrelevant text! Now, try to address the following problem using the above guidelines: